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A comparison of the behavioural responses of fishes to a remotely operated vehicle and diver-based stereo-video sampling

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ABSTRACT

Successful monitoring of fishes to inform adaptive management relies on accurate and robust data on a range of indicators including diversity, abundance, and length-based metrics. The methods for gathering this data have evolved from destructive techniques to underwater visual census (UVC), and now emergent methods such as divers or remote operated vehicles (ROVs) equipped with stereo-video systems. Stereo-video techniques can provide accurate information on traditional metrics such as abundance, length, and biomass, as well as metrics on fish behaviour. Fish wariness, measured by the minimum distance between fish and an observer or stereo-video system before the fish moves away (minimum approach distance; MAD) is an emerging metric for assessing changes in fishing pressure which can be more sensitive than traditional metrics. Differences in fish behaviour across methods can bias results and influence measurement accuracy and precision of stereo-video systems, because accuracy decreases as the distance of fish increases away from the cameras. We assessed the behaviour of six focal species towards a diver using a stereo-video system (stereo-DOV) and a stereo-video system mounted on a remotely operated vehicle (stereo-ROV). We also compared the fish lengths and the length frequency distribution of measurements from the two methods. We aimed to increase the understanding of the relative abilities of the two methods at gathering length-based data and the differences in fish behavioural responses towards the methods. There were no differences in the proportion of length measurements or the mean length, and small differences in the length frequency distribution sampled by the two methods. We found that MAD differed between the methods and the direction of this difference was species-specific. This was likely due to different trophic traits such as foraging modes and behavioural patterns between these species and potentially adaptations resulting from fishing pressure. Indicator species such as *Pomacentridae* species showed more aggressive behaviour towards SCUBA divers, likely due to defence of their territories, while fisheries targeted species such as *Choerodon rubescens* showed more wariness towards divers as a possible adaptation to fishing pressure. At the assemblage level, the relationship between MAD and fish length was weaker with the stereo-ROV, indicating that larger fish showed more wariness towards SCUBA divers than the ROV. The decreased wariness of larger fish and fisheries targeted species towards the stereo-ROV compared to stereo-DOV, as well as the similarities in the length frequency distributions sampled by the two methods suggest that stereo-ROV may provide a suitable alternative to stereo-DOV.

1. Introduction

Monitoring of marine fishes relies on a robust assessment of fish diversity, abundance, size structure and biomass (Jennings, 2004; Jennings and Kaiser, 1998; Schramm et al., 2020a). Changes in technology in the last three decades have facilitated the development of video-based methods for collecting biological data on fish assemblage composition (Cappo et al., 2003, 2006; Harvey and Shortis, 1995; Mallet and Pelletier, 2014; Schramm et al., 2020a, 2020b; Sward et al., 2019). These methods have been designed to overcome limitations such as depth

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constraints of diver-based methods or the need for *in-situ* taxonomic expertise with underwater visual census (UVC). When designing monitoring or research programs for fish, it is important to understand the limitations and biases associated with the sampling methods available (Harmelin-Vivien and Francour, 1992; MacNeil et al., 2008). For example, behavioural and physiological traits of different fishes such as wariness and trophic level may mean that different methods sample the same species with different detectability (Cappo et al., 2006; Langlois et al., 2010; Schramm et al., 2020a; Watson et al., 2010). Methods may also differ in their efficiency and cost in field and laboratory time (Goetze et al., 2015; Langlois et al., 2010), health and safety considerations, and their suitability to sample within no-take marine reserves (Cappo et al., 2003, 2006). Since the development of the initial diver operated stereo-video (stereo-DOV) prototype by Harvey and Shortis (1995), stereo-video methods have become widely used to gather data on fish assemblage compositions. The use of stereo-video facilitates accurate measurement of fish length (Harvey et al., 2002) and definition of transect area through the ability to reject observations made outside of a specified transect width or distance (Harvey et al., 2004). Stereo-video also facilitates quantitative counts of abundance (Harvey et al., 2007; Watson et al., 2005) with the ability for experts to review identifications and measurements during post processing of imagery (Cappo et al., 2006). Another advantage of stereo-video sampling is its ability to quantify fish behaviour using minimum approach distance (MAD) as a proxy for wariness towards divers (Lindfield et al., 2014; Stamoulis et al., 2020). This has been shown to be a sensitive indicator of fishing pressure in areas where spearfishing is common (Goetze et al., 2017) and therefore provides a useful tool for monitoring the effects of fishing pressure. This study compares the more widely used and accepted method of stereo-DOVs (Goetze et al., 2019) with the newer method of stereo-ROV which combines existing stereo-video technology with emerging observation class ROV technology (ROV classes defined by Monk et al. (2020)).

While consideration of a variety of assessment metrics is essential in designing monitoring programs (Nash and Graham, 2016), length-based metrics (including biomass) can be more sensitive indicators to changes in fishing pressure than abundance (Claudet et al., 2006; Taylor et al., 2014). Length data can provide insight into fishing pressures that are size selective (Nash and Graham, 2016), and non-fishing related changes such as recruitment events (Shin et al., 2005). It can also add context to the interpretation of abundance data as the functional impact of larger individuals can be greater and more widespread than smaller individuals. Furthermore, length-based metrics can be a predictor for trophic levels. This is an important metric to detect changes in trophic structure that may be caused by acute disturbances like fishing pressure (Jennings et al., 2002) and more complex disturbance regimes such as habitat degradation which arise from multiple human pressures (Taylor et al., 2022). The use of stereo-video sampling provides a means to gather data which provide accurate length based metrics. These metrics are an important consideration for adding context to abundance and species richness data and moving towards an ecosystem approach to fisheries management by interpreting functional ecological effects of changes in community structure (Jennings et al., 1995; Jennings and Polunin, 1997; Shin et al., 2005; Stuart-Smith et al., 2017).

Remotely operated vehicles (ROVs) have been tested as an alternative sampling technique for assessing fish assemblage composition (Ajemian et al., 2015; Andaloro et al., 2013; Consoli et al., 2016; Hellmrich et al., 2023; Norcross and Mueter, 1999; Trenkel et al., 2004; Vigo et al., 2023). Sward et al. (2019) present a systematic review of the literature that use ROVs as a visual survey tool to assess marine biota, highlighting the need for standardised sampling protocols and further investigation into behavioural reactions of marine fauna towards ROVs. A combination of ROV technology and stereo-video technology has more recently been developed to assess fish assemblage parameters including abundance, length, and biomass (Hellmrich et al., 2023; Jessop et al., 2022; Schramm et al., 2020a; Schramm et al., 2020b; Sward et al., 2019). This combination of technologies benefits from both the accuracy and precision of stereo-video and the advantages of remote sampling including increased health and safety and the ability to sample deeper waters (Hellmrich et al., 2023; Parry et al., 2003; Schramm et al., 2020a; Schramm et al., 2020b; Smolowitz et al., 2015; Sward et al., 2019; Warnock et al., 2016). Data collected by stereo-ROVs has been compared to other methods including stereo-DOVs, stereo-BRUVs, underwater visual census (UVC) and slow towed stereo-video with varying results (Hellmrich et al., 2023; Jessop et al., 2022; Schramm et al., 2020a; Schramm et al., 2020b; Warnock et al., 2016). Differences in assemblage composition sampled by stereo-ROV and stereo-DOV have been attributed to differences in the height of operation above the substrate (altitude) with stereo-DOV sampling more individuals and species than stereo-ROV (Schramm et al., 2020b). When attempting to pilot the stereo-ROV at the same altitude as the standardised method for stereo-DOV (~50 cm above substrate; Goetze et al., 2019), the two methods sampled a statistically similar assemblage and number of species. However, stereo-DOV still sampled more individuals than stereo-ROV (Jessop et al., 2022). This was attributed to the shorter time taken to complete transects with stereo-ROV and difficulty maintaining the 50 cm altitude with challenges such as ocean currents, swell, and highly complex reef topography (Jessop et al., 2022; Schramm et al., 2020b). Further investigation into the abilities of stereo-DOV and stereo-ROV to sample a variety of population dynamics including length, biomass and behaviour is important to understand the advantages and biases of the two methods.

Changes in behaviour of fishes may influence the accuracy and precision of length measurements, counts of abundance, and the number of species observed by a stereo-video system. For example, the accuracy and precision of length measurements of fish increases as a fish's distance from the stereo-DOV or stereo-ROV decreases (Harvey et al., 2010). Consequently, the length estimates of fish which are highly wary and remain further away from the ROV may have lower accuracy and precision. The noise of bubbles produced by SCUBA divers also change fishes behaviour. For example, an assessment using closed circuit rebreathers (CCR) recorded more species and greater abundance than conventional open circuit SCUBA outside an MPA, but the two methods recorded similar metrics inside the MPA (Lindfield et al., 2014). Difference in fish behaviour towards observation class ROVs and open circuit SCUBA divers have been observed on a tropical reef, with fish getting closer to an ROV than divers (Hellmrich et al., 2023). These behavioural effects vary with many factors including increases in fishing pressure leading to increased wariness of more highly targeted and larger species in highly fished areas (Alós et al., 2012, 2015; Gray et al., 2016; Januchowski-Hartley et al., 2015; Kulbicki, 1998). Behavioural effects can also vary with trophic traits such as foraging mode and individual traits such as movement around a fish's home range (Alós et al., 2012). The types of fishing taking place (Alós et al., 2012), and the proximity to a marine protected area (MPA) can also impact fish behaviour (Alós et al., 2015; Januchowski-Hartley et al., 2014). The potential of different sampling techniques to influence fish behaviour is an important consideration for designing monitoring programs with visual methods.

At a species-specific level, fish behaviour has been shown to vary across life stages, with larger individuals showing more wariness than smaller individuals (Goetze et al., 2017). An understanding of the behaviour of fishes towards ROVs has previously been limited to a few species-specific studies involving work class ROVs (Laidig et al., 2013; Lorance and Trenkel, 2006; McLean et al., 2017; Ryer et al., 2009; Stoner et al., 2008; Trenkel et al., 2004), which are larger and emit more light and sound than observation class ROVs, potentially inflating behavioural effects (Sward et al., 2019). More recently, species-specific responses towards an observation class ROV have been investigated in a tropical location, with fish generally being less wary of the ROV compared to divers (Hellmrich et al., 2023). The use of observation class ROVs for sampling fish may reduce some of the behavioural bias that is

exhibited by fish towards divers and work class ROVs, because observation class ROVs are small and do not produce bubbles (Stamoulis et al., 2020). Species-specific responses to stationary SCUBA divers with open circuit systems have also been recorded. Some species have shown avoidance of divers with observed fish density declining by up to 77% in the presence of divers (Stanley and Wilson, 1995), while other species, such as *Coris auricularis* have shown attraction to divers (Watson and Harvey, 2007), possibly due to an association of diver presence with food availability from seabed disturbance (Chapman et al., 1974). Species-specific behaviour, across a range of species and fishing pressure is a useful metric in comparing stereo-DOV and stereo-ROV as a reduction in avoidance may allow for a higher detectability across life stages and trophic groups, and a higher degree of measurement accuracy.

We assessed differences in length data and the behavioural interactions of fish with SCUBA divers and observation class ROVs. Specifically, we aimed to compare (1) the number of length measurements that could be made from imagery collected using stereo-DOV and stereo-ROV, (2) the length frequency distribution of the fish measurements gathered by each method, (3) the wariness of fish (MAD) to each method, and (4) the relationship between MAD and fish length with each method.

2. Materials & methods

2.1. Study area

Sampling was conducted in the Jurien Bay Marine Park (JBMP) during April and May of 2021. The JBMP covers over 82,000 ha of the nearshore ocean, adjacent to the central west coast bioregion of Western Australia (Fig. 1). All sampling took place in the lagoon area at depths ranging between 4 and 10 m. The area lies within a limestone reef system in part of the Great Southern Reef (Bennett et al., 2015). It hosts a unique assemblage of tropical and temperate marine species (Department of Conservation and Land Management, 2005). The marine flora in the JBMP is historically dominated by *Ecklonia radiata* (kelp) canopy and a diverse assemblage of foliose algae in reef areas intertwined with seagrass meadows (Wernberg et al., 2006). The JBMP also hosts a community of coral species supported by warm water flowing poleward with the Leeuwin Current, coral cover has been estimated at up to 30% benthic cover in some areas (Ross et al., 2021). The fish assemblage associated with this reef system is diverse, and this study investigates the behaviour of the most abundant groups in this system including the large schooling *kyphosids* (drummers), carnivorous *labrids* (wrasse), and large territorial *pomacentrids* (damselfish). Zoning in the JBMP includes general use; which allows both recreational and commercial fishing, sanctuary zones (fully protected areas); where all extractive activities are prohibited and special purpose (scientific reference zones); that allow commercial and recreational fishing for rock lobster only (Fig. 1).

2.2. Experimental design

A comparison of stereo-DOV and stereo-ROV was completed at nine sites, each site was separated by at least 250 m (Fig. 1). At each site, six 50 m belt transects were completed for each method, and all transects were completed in a single dive for both the stereo-DOVs and stereo-ROVs. All transects were 5 m wide and 5 m high and had a separation of ≥10 m. Transect width was maintained using 'EventMeasure Stereo' (https://www.seagis.com.au/event.html) to reject any observations greater than 2.5 m either side of the middle of the transect. This width was chosen to be small enough to minimise changes in detectability at the far edges of the transect. Consistency was maintained between the two methods by sampling in similar visibility and lighting conditions and transects were all completed with the sun behind the camera system to minimise backlighting. We confirmed that there were not differences in the water visibility conditions between the two methods using a *t*-test

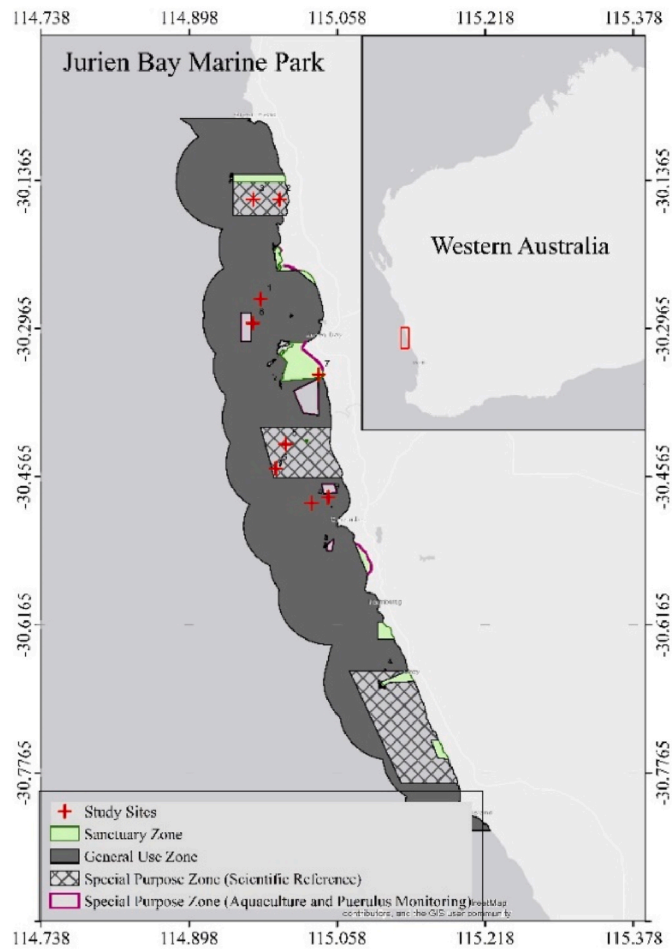

Fig. 1. Map of survey location, highlighting sample sites along the central west coast of Western Australia in the Jurien Bay Marine Park.

after checking the data for homogeneity of variance. Management zoning was not considered as a factor in the analysis as previous studies detected no difference in finfish assemblage across zones in the JBMP (Jessop et al., 2022) and we were unable to sample across the zones with sufficient power due to time and weather constraints.

2.3. Stereo-video systems

The stereo-video systems consisted of two cameras, mounted on a base bar at a slight inward converging angle, as described by Harvey and Shortis (1995, 1998). The separation of the cameras was 700 mm with an inward converging angle of 8° for the stereo-DOV and 595 mm with an inward converging angle of 5° for the stereo-ROV.

2.3.1. Diver operated stereo-video (stereo-DOV)

Stereo-DOV surveys were completed with two SCUBA divers using conventional open circuit systems, following the method outlined by Goetze et al. (2019). On average, transects were completed in 1 min and 44 s ±0.041 (1SE). The cameras used on the stereo-DOV system were Sony FDR X3000 action cameras recording at 1080p, 60 fps, and a medium field of view to reduce motion parallax associated with the stereo video system and fish moving simultaneously.

2.3.2. Stereo-video remotely operated vehicle (stereo-ROV)

The stereo-ROV used was an observation class Blue Robotics Blue-ROV 2 (457 × 330 × 254 mm) that was fitted with a stereo-video system with two Sony RX0 II video cameras recording at 1080p and 50 fps. The stereo-ROV unit was operated approximately 50 m ahead of a surface

vessel and was attached by an umbilical control cord. The vessel maintained this distance while the stereo-ROV moved along the transect. Its location was tracked using a Seatrac X150 USBL Beacon at the boat and X010 Modem Beacon attached to the stereo-ROV. The USBL system had a range resolution of ±0.1 m and angular resolution of approximately two percent of the acoustic range (approximately 1 m at the range used in this study). The USBL facilitated navigation, but also allowed us to calculate transect length during post processing by matching the known time to the known location of the ROV (see supplementary data for code used). Where possible, the stereo-ROV was flown approximately 50 cm above the benthos to match the stereo-DOV procedure (Goetze et al., 2019). This was not always possible in complex reef where sudden changes in topography occurred. The footage from each site was separated into transects of the specified 50 m length from the tracking data from the USBL, on average each transect took 1 min and 26 s ±0.041 (1SE).

2.4. Calibration and image analysis

Calibration of each stereo-video system was completed before and after the sampling. The software 'CAL' (https://www.seagis.com.au/bundle.html) was used, following the methods outlined by Harvey and Shortis (1998). Video analysis was completed using the software 'EventMeasure Stereo'. For the transect methods, all fish within the specified transect boundaries were identified to the lowest possible taxonomic level (usually species) and fork length (from the tip of the snout to the fork of the caudal fin) was measured. Where a length measurement was not able to be recorded due to part of the fish being obscured in the field of view, a 3D point was used to record abundance and MAD data. Where a species level identification was not possible, individuals were identified to genus. For MAD analysis, all fish within transects were measured when they were closest to the camera system, following the protocol outlined by Goetze et al. (2019). All observations of fish were limited to within a range of 7 m from the cameras to maintain accuracy and precision (Harvey et al., 2010). Any measurements with a residual mean square (RMS: an estimator of the quality of measurement (Harvey and Shortis, 1998)) greater than 20 mm or precision greater than 10% of the fork length were rejected as per guidelines (Harvey et al., 2010).

2.5. Statistical analysis

Statistical analysis was conducted using the programming language R in R studio (Version: RStudio, 2022.07.1 + 554 "Spotted Wakerobin" Release, R Core Team, 2022). Analysis was performed at the pooled assemblage level and on focal species. These focal species were defined as species which had a sample size of greater than 25 for both stereo-DOV and stereo-ROV to ensure confidence in the regression model fit (Jenkins and Quintana-Ascencio, 2020). For the MAD and length frequency analysis, stereo-DOV and stereo-ROV models were computed as separate models but displayed on the same axis to allow for comparisons.

2.5.1. Length frequency analysis

The relative ability of stereo-DOV and stereo-ROV to measure length was compared using the proportion of length measurements to the number of fish detected as a proxy. A two-sample *t*-test assuming equal variance (R Core Team, 2022) was used on the percentage of observations where a length measurement was made. Homogeneity of variance was tested using a Levene's test. A permutational analysis of variance (PERMANOVA) was used to test for differences in the mean length between the methods using Primer 7 with the PERMANOVA + add on (Anderson et al., 2008). This test was used because the length data violated the assumption of normality.

Kernel density estimates (KDE) were constructed for the whole assemblage following the methods published by Langlois et al. (2012) using the packages KernSmooth (Wand, 2021) and sm (Bowman and Azzalini, 2021). Tests for both shape and location of the distributions were done by comparing the area between the KDEs of the two methods to the area resulting from permutations of the data into random pairs. This was represented graphically with a grey band centred on the mean KDE ±1 SE and based on a null model of no difference (Langlois et al., 2012). Assemblage level comparison of length frequency distribution sampled by the two methods was justified because the species composition of the assemblage sampled by stereo-DOV and stereo-ROV was not significantly different (Jessop et al., 2022). This comparison was used to test for any differences in the detectability of fishes a range of lengths between the two methods. Large schools of *Neatypus obliquus* and *Kyphosus sydneyanus* that were encountered at one site with stereo-DOV and not with the stereo-ROV were removed from subsequent analysis due to their disproportionate effect on the data. Separate KDE plots were constructed both with and without these schools of fish to demonstrate the effect of their removal.

2.5.2. Minimum approach distance

To assess the relationship between MAD and fork length, linear models were created using R base statistics (R Core Team, 2022) and plotted using ggplot2 (Wickham, 2016). This was done at an assemblage level (with *N. obliquus* and *K. sydneyanus* removed) and for focal species. Fish less than 5 cm were removed from the analysis as they could not be measured with a high degree of accuracy across the full range (distance from camera out to 7 m) (Goetze et al., 2017; Harvey et al., 2010). Regression lines were plotted with confidence intervals and adjusted R^2 values were calculated. An ANOVA design consisting of one factor with two levels (method, stereo-DOV and stereo-ROV) and length as a covariate with MAD as the response variable was used to test the relationship between MAD and method and MAD and fish length. The R package emmeans (Lenth, 2023) was used to test for significant differences in the slopes of the regressions between the two methods. Pirate plots were created in the package ggplot2 (Wickham, 2016) to visualise the relationship between MAD and method at an assemblage and species-specific level.

3. Results

A total of 8,399 observations of fish were made across the two methods. Of these observations 4,933 were made by stereo-DOV and 3,466 were made by stereo-ROV. A total of 1,536 length measurements and 3,397 3D points were recorded by stereo-DOV and a total of 2,091 length measurements and 1,375 3D points were recorded by stereo-ROV. *Coris auricularis* was the most abundant species with 2,229 observations by stereo-DOV and 1,799 observations by stereo-ROV. Six species had greater than 25 observations by both methods and were used as focal species. We found no significant difference in the visibility conditions between the two methods ($t = -0.197$, $df = 16$, $p = 0.846$), the mean visibility encountered was 6.95 ± 0.539 m and 7.11 ± 0.550 m for stereo-DOV and stereo-ROV respectively, the minimum visibility encountered across all methods was 4.22 m.

3.1. Length frequency analysis

At the assemblage level there was no significant difference in the proportion of the total number of fish which could be measured for length using stereo-DOV or stereo-ROV ($t = -1.61$, $df = 16$, $p = 0.127$). Although, the mean proportion of successful length measurements for stereo-DOV was lower at 27.0% ± 4.23 compared to 36.1% ± 3.77 for stereo-ROV. There was also no significant difference in the mean length of fish observed by the two methods (Method, Pseudo-$F_{(1, 2623)} = 0.659$, $P = 0.428$) with stereo-DOV recording a mean length of 113 ± 2.40 mm (1SE) and stereo-ROV 128 ± 2.20 mm (1SE).

The length frequency distributions observed by the two methods were both positively skewed with similar modes (Fig. 2). The shape

differed as stereo-DOV had a bimodal distribution and stereo-ROV had a higher probability of observing individuals at the lower mode than stereo-DOV. The areas where the two methods differed significantly were illustrated by the regions where the distributions fell outside of the null model of no difference represented by the grey shaded area (Fig. 2). The second mode in the stereo-DOV data consisted largely of observations of a school of *Neatypus obliquus* at one site. Stereo-DOV also observed a group of fishes with a fork length between 600 mm and 800 mm that was absent in stereo-ROV observations. These individuals were all from a school of *Kyphosus sydneyanus* which was observed at one site. The statistical test for location of the two distributions also showed a significant difference in the length frequency distribution observed by both stereo-DOV and stereo-ROV (Fig. 2). Removal of the schools of both *K. sydneyanus* and *N. obliquus* increased the similarity of the shape and location of the length frequency distribution observed by the two methods (Fig. 2). The higher probability of stereo-ROV observing individuals at the lower mode and the bimodal distribution of stereo-DOV remained, but was less evident. The shape and location of the distributions remained statistically different between the two methods (Fig. 2).

3.2. Minimum approach distance

Linear regression plots were used to test if length was a predictor of MAD for the whole assemblage and chosen focal species for both stereo-DOV and stereo-ROV. At the assemblage level, both methods had a positive linear relationship with MAD increasing as fish length increased (Fig. 3: stereo-DOV slope coefficient = 0.39, stereo-ROV slope coefficient = 0.21, $p < 0.001$). The slopes were significantly different between the two methods ($t = 6.15$, $df = 2584$, $p < 0.001$). The analysis of variance showed there was a significant interaction between length and method and significant main effects of both length and method on MAD (Table 1). On average, stereo-DOV sampled individuals at a closer distance (mean MAD = 1980 ± 18.5 mm (1SE)) than stereo-ROV (mean MAD = 2190 ± 18.9 mm (1SE)) and MAD increased with fork length for both methods.

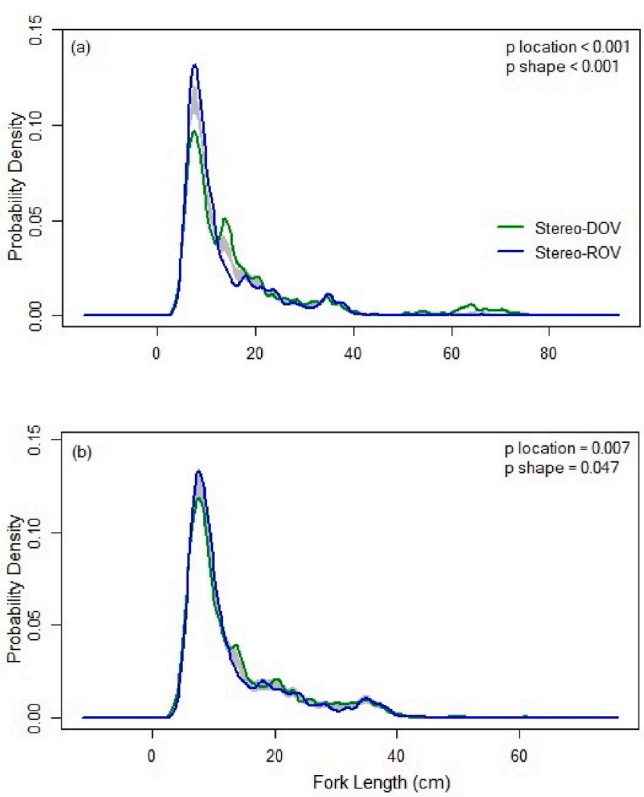

Fig. 2. Comparison of kernel density estimate probability functions for the (a) assemblage of fishes and (b) the assemblage of fishes minus schools of *Kyphosus cornelii* and *Neatypus obliquus* which were sampled by only stereo-DOV or stereo-ROV respectively. Grey band shows the null model of no difference between the two functions plus or minus one standard error. P values are shown for the significance tests assessing similarity between the shape and location of models for each method.

3.2.1. Focal species

Minimum approach distance was highly variable across the different focal species with all species except the non-targeted *Kyphosus cornelii* showing a significant increase in MAD with an increase in fork length (Table 1, Fig. 3). There was a significant effect of method on MAD for all species (Table 1), however the direction of this difference varied among species (Fig. 3). Stereo-ROV sampled *K. cornelii* and *Choerodon rubescens* at a significantly lower MAD than stereo-DOV. Stereo-DOV sampled all other focal species (*Coris auricularis*, *Notolabrus parilus*, *Parma mccullochi* and *Pomacentrus milleri*) at a significantly lower MAD than stereo-ROV (Table 1, Fig. 3). *Coris auricularis* was the only species for which the relationship between MAD and length significantly differed between the two methods ($t = 2.35$, $df = 1610$, $p = 0.019$), the slope of the regression for stereo-ROV was more gradual (slope coefficient = 0.039) than for stereo-DOV (slope coefficient = 0.052 (Fig. 3). All other focal species had no significant difference in the slopes of the regressions between the two methods ($p > 0.05$). *Choredon rubescens* appeared to have a steeper slope for stereo-DOV (slope coefficient = 0.032) compared to stereo-ROV (slope coefficient = 0.006), however, this difference was not statistically significant at $\alpha = 0.05$ ($t = 1.81$, $df = 55$, $p = 0.076$) (Fig. 3). *Kyphosus cornelii* and the two *Pomacentridae* species (*P. mccullochi* and *P. milleri*) had low adjusted R^2 values fitted to the regression line of fork length versus MAD (adjusted $R^2 \leq 0.082$).

4. Discussion

Our comparison of stereo-DOV and stereo-ROV found that at the assemblage level, larger fish were less wary of the stereo-ROV than the stereo-DOV. Previous studies have observed a consistent relationship with larger individuals being more wary of divers than smaller individuals (Goetze et al., 2017; Hellmrich et al., 2023; Januchowski-Hartley et al., 2014; Januchowski-Hartley et al., 2015). Hellmrich et al. (2023) found that overall, fish sampled on a tropical reef were less wary of an observation class ROV compared to SCUBA divers, and that larger, more frequently fished species had a steeper relationship between MAD and fork length that smaller fish. The reduced wariness of fish towards the stereo-ROV may increase the measurement accuracy for larger fish when compared to stereo-DOV. This is because measurement accuracy decreases with increasing distance (Harvey et al., 2010). The length frequency distribution of fishes measured by stereo-DOV and stereo-ROV were broadly comparable, and ROV sampling resulted in a slightly higher percentage of successful length measurements. While we observed some differences in the length frequency distribution sampled by the two methods, they were primarily driven by chance encounters with large schools and the differences were negligible once these schools were removed from the analysis. Overall, the number of length measurements was low due to moderate visibility in the water at the time of the survey and kelp obscuring fish in the field of view of one of the cameras. Jessop et al. (2022) found that stereo-DOV and stereo-ROV were also comparable in the assemblage composition and number of species sampled with some differences in the number of individuals sampled. We provide further evidence to support the use of stereo-ROV for reef fish sampling in monitoring programs where length data is required. Our data also indicates that stereo-ROVs may outperform stereo-DOVs where measuring large fish and targeted species is important.

The technological differences between stereo-DOV and stereo-ROV, including emission of artificial noise by the thrusters or electronics pod on the ROV, and the bubbles emitted by SCUBA divers, provide a

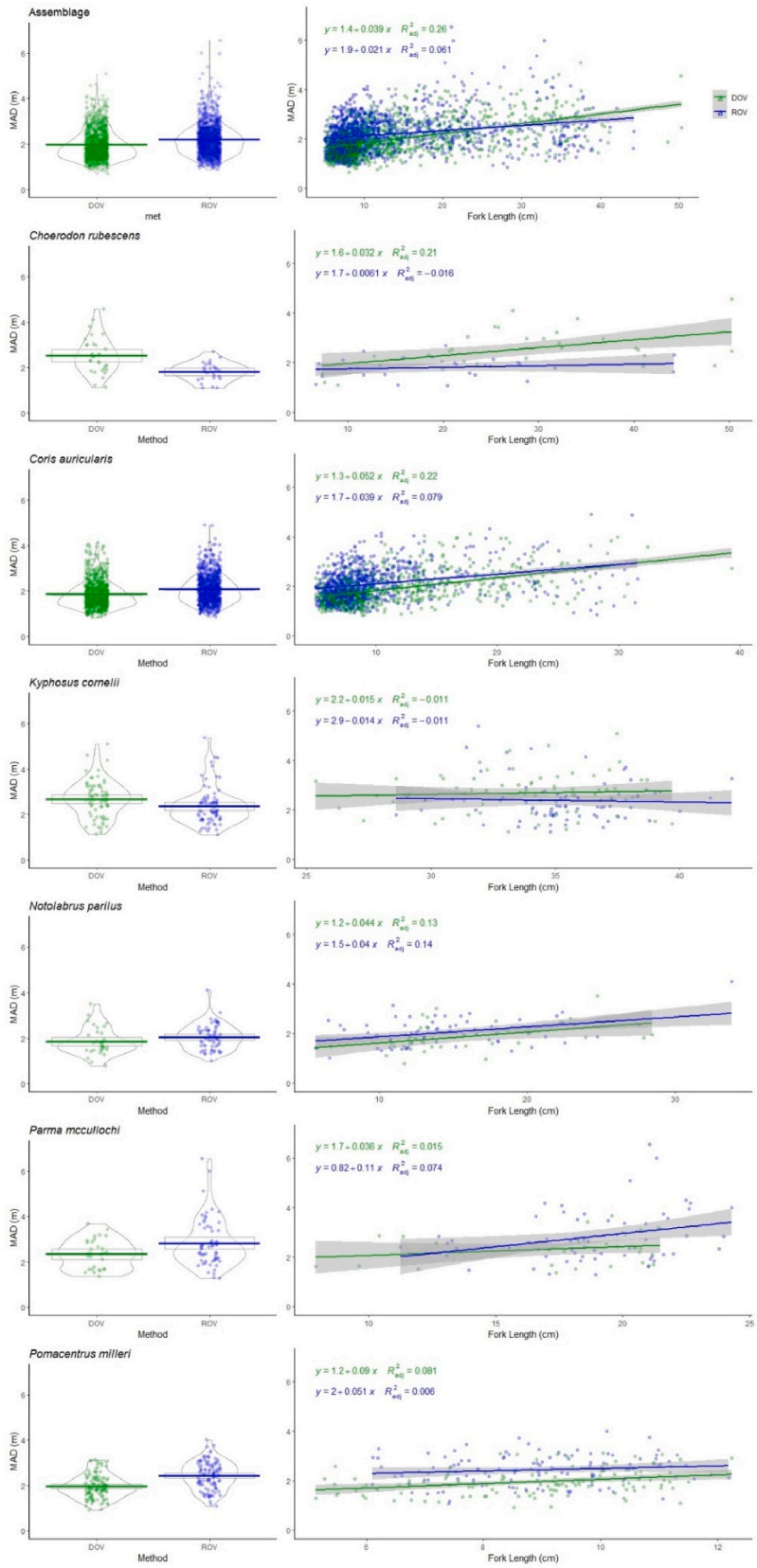

Fig. 3. Assemblage level and species-specific pirate plots (column 1) and linear regression plots (column 2) illustrating the effect of method on MAD. Pirate plots show data as points, a horizontal-coloured line indicating mean, boxes indicating 95% confidence intervals, and violins indicating density distributions. Linear regression plots show data as points with a fitted linear regression line and 95% confidence interval.

Table 1
Analysis of co-variance assessing the effect of fish length and method on the minimum approach distance of fish. Values in bold show statistical significance at α = 0.05.

	Assemblage						*Choerodon rubescens*				
	df	Sum Sq	Mean Sq	F value	p		df	Sum Sq	Mean Sq	F value	p
Length	1	157.08	157.08	405.41	**< 0.001**	Length	1	7.05	7.05	20.13	**< 0.001**
Method	1	34.65	34.65	89.43	**< 0.001**	Method	1	4.02	4.02	11.47	**0.001**
Length × Method	1	14.64	14.64	37.78	**< 0.001**	Length × Method	1	1.14	1.14	3.26	0.076
Residuals	2584	1001.20	0.39			Residuals	55	19.27	0.35		
	Coris auricularis						*Kyphosus cornelii*				
	df	Sum Sq	Mean Sq	F value	p		df	Sum Sq	Mean Sq	F value	p
Length	1	77.19	77.19	240.48	**< 0.001**	Length	1	0.17	0.17	0.25	0.619
Method	1	30.02	30.02	93.52	**< 0.001**	Method	1	3.17	3.17	4.69	**0.032**
Length × Method	1	1.78	1.78	5.53	**0.019**	Length × Method	1	0.25	0.25	0.37	0.545
Residuals	1610	516.82	0.32			Residuals	143	96.50	0.67		
	Notolabrus parilus						*Parma mccullochi*				
	df	Sum Sq	Mean Sq	F value	p		df	Sum Sq	Mean Sq	F value	p
Length	1	4.38	4.38	16.11	**< 0.001**	Length	1	6.47	6.47	7.88	**0.006**
Method	1	1.15	1.15	4.24	**0.042**	Method	1	3.33	3.33	4.06	**0.047**
Length × Method	1	0.01	0.01	0.04	0.850	Length × Method	1	1.21	1.21	1.48	0.227
Residuals	95	25.82	0.27			Residuals	87	71.41	0.82		
	Pomacentrus milleri										
	df	Sum Sq	Mean Sq	F value	p						
Length	1	3.61	3.61	12.04	**0.001**						
Method	1	12.87	12.87	42.92	**< 0.001**						
Length × Method	1	0.21	0.21	0.71	0.401						
Residuals	224	67.19	0.30								

possible explanation for the observed differences in MAD. We found species specific patterns of fish behaviour towards divers and the stereo-ROV with species such as the Western King Wrasse (*Coris auricularis*) and the Brownspotted Wrasse (*Notolabrus parilus*) appearing to be attracted towards divers, indicated by the lower MAD for stereo-DOV compared to stereo-ROV. The Western King Wrasse was the only species to have a significant difference in the slope of the MAD – length relationship between the two methods, meaning that larger fish of this species were more wary of the DOV than the ROV. However, the strength of these relationships was relatively low with both methods, but may have been statistically significant due to the large sample size for this species increasing the ability to detect small differences (n stereo-DOV = 801, n stereo-ROV = 839). The potential attraction towards divers may be due to the presence of bubbles, noise, or the association of divers with food available from disturbing the seabed (Chapman et al., 1974). These two wrasse species are carnivorous and feed on benthic invertebrates (Vanderklift et al., 2007) which supports the hypothesis that they are attracted to divers due to food availability associated with divers disturbing the seabed (Chapman et al., 1974). This pattern is likely to vary across locations which have been subject to different human impacts over time. Fish may be attracted to or avoid the bubbles and the noise of SCUBA divers (Gray et al., 2016; Lindfield et al., 2014; Watson and Harvey, 2007), or the noise and light associated with ROVs (Logan et al., 2017; Ryer et al., 2009; Stoner et al., 2008). In the case of the Western King Wrasse, any attraction towards divers may not be consistent throughout the fish's life stages, as larger fish were slightly more wary toward the DOV than smaller fish. This increased wariness was not as apparent towards the ROV. The noise levels have been quantified for SCUBA divers (Radford et al., 2005), however the noise range of observation class ROVs has not been measured. This topic warrants further investigation in understanding the magnitude of the influence of sound produced by observation class ROVs on the behaviour of fishes.

Other species-specific behavioural responses were observed, which could be related to the trophic traits and foraging modes of the species, or levels of fishing pressure (Alós et al., 2012, 2015; Januchowski-Hartley et al., 2014). For example, Baldchin Groper (*Choerodon rubescens*) is a highly targeted species in the JBMP (Ryan et al., 2019) and an indicator species for the region. Baldchin Grouper displayed less wariness towards the stereo-ROV compared to divers and had a significant relationship between fork length and MAD across both methods. The relationship between fork length and MAD for Baldchin Grouper did not significantly differ between the two methods. However, when displayed graphically, there appeared to be an interaction, with larger fish being more wary of divers than the ROV. We hypothesise that this was likely due to a small sample size of this species (n stereo-DOV = 33, n stereo-ROV = 28) and suggest that this pattern be further investigated. Our analysis indicates that larger Baldchin Groper were likely less wary of the stereo-ROV which could improve the measurement accuracy of these fish. Conversely, the Western Buffalo Bream (*Kyphosus cornelii*) which is not targeted by fishers, had very weak, shallow relationship between fork length and MAD for both stereo-ROV and stereo-DOV. This is consistent with previous work which has suggested that larger targeted fish are more wary compared to non-targeted species and that observation class ROVs generally have a smaller MAD than divers (Hellmrich et al., 2023). It is possible that the differences in Baldchin Groper behaviour between the methods may be due to fishing pressure which has been shown to influence targeted fish behaviour (Bergseth et al., 2016; Goetze et al., 2017; Januchowski-Hartley et al., 2014). The JBMP has relatively low levels of spearfishing pressure compared to line fishing (Department of Conservation and Land Management, 2005), however, anecdotally spearfishing of species including Baldchin Grouper does occur (Harvey pers. obs.). The influence of management zoning on fish behaviour was not accounted for in this study due to insufficient replication inside and outside of protected areas. However, Hellmrich et al. (2023) demonstrated that fish were less wary of divers and ROVs inside tropical MPAs compared to outside. We recommend further investigation to test the magnitude of effects of spearfishing pressure on behaviour towards divers and observation class ROVs across a wider range of ecosystems and levels of fishing. In areas of higher spearfishing pressure, it has been argued that observation class stereo-ROVs may be a more suitable technique than stereo-DOVs due to the behavioural responses of targeted species (Hellmrich et al., 2023). A potential dampening of this response when sampling with observation class stereo-ROVs could increase the number of larger individuals and targeted species being observed and improve both the accuracy of length measurements and the proportion of length measurements to 3D point observations. Conversely, this may reduce the ability to use MAD as an indicator for spearfishing pressure when sampling with stereo-ROV. Previous studies found the wariness of targeted fish towards divers increased because of spearfishing fishing pressure (Bergseth et al., 2016;

Goetze et al., 2017; Januchowski-Hartley et al., 2014). This behavioural avoidance may result in difficulty measuring targeted species with stereo-DOV in areas of high spearfishing pressure.

We also observed differences in behaviour of fishes that appeared to be due to their foraging mode. Both the McCulloch's Scalyfin (*Parma mccullochi*) and Miller's Damselfish (*Pomacentrus milleri*) are members of the *Pomacentridae* family and had a smaller MAD with stereo-DOV than stereo-ROV as well as weak relationships between fork length and MAD. Like other territorial damselfish, these species maintain a patch of food algae by weeding undesirable algae species and agonistically excluding other herbivores from their patch (Ceccarelli et al., 2001; Saunders et al., 2015; Shalders et al., 2018). As a result, these species approach and demonstrate aggressive behaviour towards divers as they defend their territory. The larger MAD observed with stereo-ROV may be a result of these species not perceiving the ROV as a threat to their territory, and therefore not exhibiting defensive behaviours. This finding supports other literature that has found behaviour of fishes towards ROVs to be species specific (Hellmrich et al., 2023; Laidig et al., 2013; Lorance and Trenkel, 2006) and highlights the importance of understanding the advantages and biases of ecological sampling techniques which may vary by geographical location and ecosystem type.

When considering the behavioural effects that we have observed, it is important to highlight that our observations have been limited to those that are within the visual depth of field of underwater cameras. Uncertainty remains regarding species and individuals that may exhibit avoidance of both methods to the degree that they are not detectable. This is likely occurring at some level as fish have been shown to be able to detect the sound emitted by SCUBA divers by up to 200 m away (Radford et al., 2005). Further investigation of behaviour beyond the visual range could be done with an ROV using a combination of stereo-video and multibeam sonar. This combination of technology would be a unique and valuable approach to furthering our understanding of fish behavioural responses towards both observation class ROVs and SCUBA divers.

We aimed to operate both the stereo-DOV and stereo-ROV ~50 cm above the benthos as defined in the stereo-DOV standard operating procedure (Goetze et al., 2019). However, it was difficult to maintain this altitude with stereo-ROV. Stereo-DOV sampling may also be subject to a lack of consistency in operating altitude. However, the issue is likely to be worse for stereo-ROV due to complex topography increasing the risk of possible cable entanglement with reef structure, and the decreased visual capacity of a ROV pilot viewing live camera footage compared to a diver. This is an important aspect of stereo-ROV sampling as it is thought to impact the number of individuals and species sampled (Schramm et al., 2020b). We recommend the use of sonar technology to track or lock operating altitude. The standardisation of stereo-ROV sampling altitude would be useful in improving standardisation of sampling which is beneficial for temporal and cross system comparisons (Bax et al., 2019). Minor adjustments may need to be made in different marine ecosystems to accommodate for unique ecosystem parameters, for example, altitude may need to be increased in areas with large branching kelp. This warrants further investigation of the abilities of these methods across tropical and temperate ecosystems.

The ability of a method to gather accurate data on fish length is important because length-based metrics are a sensitive indicator to changes in fishing pressure (Claudet et al., 2006). Length data also provides insight into size selective fishing pressures (Nash and Graham, 2016) and adds context to interpretation of abundance data (Lokrantz et al., 2008). Another application of this length data is the use of size-spectrum models which can be used to predict changes in size-spectra through time and to compare this among areas subject to different levels stressors such as fishing, climate disturbances, and pollution. This may assist in understanding how fishing pressure is affecting ecosystem function in conjunction with various co-occurring environmental changes (Blanchard et al., 2017). The use of stereo-ROV to gather accurate length data with a standardised methodology may allow this modelling to occur at a greater spatial scale. The proportion of length measurements gathered in this study was not significantly different between stereo-DOV and stereo-ROV at 27% and 36% respectively, but it was lower than in other studies. For example, Wilson et al. (2018) gathering ~40% with stereo-DOV and Hellmrich et al. (2023) measured between ~68% and ~70% of fish counted with stereo-DOV and stereo-ROV respectively. This was likely due to the complex canopy of kelp present in the habitat in the JBMP which sometimes obscured fish in the cameras field of view, as well as moderate visibility presenting a challenge in obtaining successful length measurements. This highlights the need to consider the advantages and biases of methods that may differ by geographical location and ecosystem type, and the importance of monitoring programs to be flexible around environmental conditions such as visibility, swell, and wind (Jessop et al., 2022).

Overall, the similarities in sampling abilities of stereo-ROV to the more widely used and accepted stereo-DOV (Goetze et al., 2015; Holmes et al., 2013; Watson et al., 2010), indicates that the use of observation class stereo-ROV is a viable alternative to stereo-DOV for sampling shallow water reef fishes. The relatively low cost of observation class ROVs compared to work class ROVs (Monk et al., 2020), provide further reasoning for the use of observation class ROVs for smaller scale research or projects with tighter budgetary constraints. This study investigated these trends in a transitionary reef system and investigation should be done across a range of tropical and temperate ecosystems to expand our understanding of the abilities and biases of the two methods, such as the assessment on a tropical reef done by Hellmrich et al. (2023). Also, consideration should be given to the unique characteristics of fishes within a specific ecosystem and potential attraction or repulsion to divers from fishing pressure, feeding of fishes, or a fish's foraging mode when selecting and interpreting data from both stereo-ROV and stereo-DOV. We believe that a larger sample size of focal species (both targeted and non-targeted) would be beneficial to further investigate the species-specific behaviour towards observation class ROVs that we have presented. Due to the logistical, safety and financial advantages of stereo-ROV over stereo-DOV and the similarities in assemblage (Hellmrich et al., 2023), number of species (Jessop et al., 2022) and length frequency sampled with each method, we recommend the use of observation class stereo-ROVs as a transect sampling method for monitoring fish assemblages. The potential dampened behavioural responses of targeted fish towards observation class ROVs compared to open circuit SCUBA divers has been investigated on a tropical reef with similar results (Hellmrich et al., 2023). Within the context of recent literature we suggest that observation class ROVs with stereo-video offer an increased ability to accurately measure and identify larger, targeted fish species compared to divers. This study supports the efficacy of the stereo-ROV sampling method for reef fish assemblages.

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CRediT authorship contribution statement

Sarah A. Jessop: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Benjamin J. Saunders:** Writing – review & editing, Visualization, Supervision, Investigation, Formal analysis, Conceptualization. **Jordan S. Goetze:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Neville S. Barrett:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis. **Euan S. Harvey:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding

acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sarah A Jessop reports administrative support and travel were provided by Western Australia Department of Biodiversity Conservation and Attractions. Jordan S Goetze reports a relationship with Western Australia Department of Biodiversity Conservation and Attractions that includes: employment.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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